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The untold story of the ‘circle of trust' behind the world's first gene-edited babies

How regulations and the scientific community itself failed to stop the “rogue” Chinese scientist He Jiankui

1 AUG 2019 • BY JON COHEN

He Jiankui, at a Hong Kong meeting in November 2018 where he presented his work, has not been seen in public since then. ANTHONY KWAN/BLOOMBERG/GETTY IMAGES

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On 10 June 2017, a sunny and hot Saturday in Shenzhen, China, two couples came to the Southern University of Science and Technology (SUSTech) to discuss whether they would participate in a medical experiment that no researcher had ever dared to conduct. The Chinese couples, who were having fertility problems, gathered around a conference table to meet with He Jiankui, a SUSTech biophysicist. Then 33, He (pronounced "HEH") had a growing reputation in China as a scientist-entrepreneur but was little known outside the country. "We want to tell you some serious things that might be scary," said He, who was trim from years of playing soccer and wore a gray collared shirt, his cuffs casually unbuttoned.

He simply meant the standard in vitro fertilization (IVF) procedures. But as the discussion progressed, He and his postdoc walked the couples through informed consent forms that described what many ethicists and scientists view as a far more frightening proposition. Seventeen months later, the experiment triggered an international controversy, and the worldwide scientific community rejected him. The scandal cost him his university position and the leadership of a biotech company he founded. Commentaries labeled He, who also goes by the nickname JK, a "rogue," "China's Frankenstein," and "stupendously immoral."

But that day in the conference room, He's reputation remained untarnished. As the couples listened and flipped through the forms, occasionally asking questions, two witnesses—one American, the other Chinese—observed. Another lab member shot video, which *Science* has seen, of part of the 50-minute meeting. He had recruited those couples because the husbands were living with HIV infections kept under control by antiviral drugs. The IVF procedure would use a reliable process called sperm washing to remove the virus before insemination, so father-to-child transmission was not a concern. Rather, He sought couples who had endured HIV-related stigma and discrimination and wanted to spare their children that fate by dramatically reducing their risk of ever becoming infected.

He, who for much of his brief career had specialized in sequencing DNA, offered a potential solution: CRISPR, the genome-editing tool that was revolutionizing biology, could alter a gene in IVF embryos to cripple production of an immune cell surface protein, CCR5, that HIV uses to establish an infection. "This technique may be able to produce an IVF baby naturally immunized against AIDS," one consent form read.

The couples' children could also pass the protective mutation to future generations. The prospect of this irrevocable genetic change is why, since the advent of CRISPR as a genome editor 5 years earlier, the editing of human embryos, eggs, or sperm has been hotly debated. The core issue is whether such germline editing would cross an ethical red line because it could ultimately alter our species. Regulations, some with squishy language, arguably prohibited it in many countries, China included.

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Yet opposition was not unanimous. A few months before He met the couples, a committee convened by the U.S. National Academies of Sciences, Engineering, and Medicine (NASEM) concluded in a well-publicized report that human trials of germline editing "might be permitted" if strict criteria were met. The group of scientists, lawyers, bioethicists, and patient advocates spelled out a regulatory framework but cautioned that "these criteria are necessarily vague" because various societies, caregivers, and patients would view them differently. The committee notably did not call for an international ban, arguing instead for governmental regulation as each country deemed appropriate and "voluntary self-regulation pursuant to professional guidelines."

Back in Shenzhen, both couples agreed to volunteer. He's study was up and running and would enroll six other couples. It proceeded quietly until the news broke in late November 2018, days before the second international summit on genome editing, in Hong Kong, China, that a couple in the trial had given birth to twin girls who had been edited while embryos. Scientists and ethicists excoriated He's medical rationale for the experiment and worried that it unnecessarily put the girls at risk. And people in the field confronted an uncomfortable truth: that regulations and the scientific community's efforts to control CRISPR's powers had failed.

A rogue scientist?

Was that because He hid his plans and deceived his colleagues and superiors, as many people have asserted? A preliminary investigation in China stated that He had forged documents, "dodged supervision," and misrepresented blood tests—even though no proof of those charges was released, no outsiders were part of the inquiry, and He has not publicly admitted to any wrongdoing. (CRISPR scientists in China say the He fallout has affected their research.) Many scientists outside China also portrayed He as a rogue actor. "I think there has been a failure of self-regulation by the scientific community because of a lack of transparency," virologist David Baltimore, a Nobel Prize–winning researcher at the California Institute of Technology (Caltech) in Pasadena and co-chair of the Hong Kong summit, thundered at He after the biophysicist's only public talk on the experiment.

Nobel laureate David Baltimore (center right) strongly denounced He Jiankui's experiment at a genome-editing summit in Hong Kong, China. ISAAC LAWRENCE/AFP/GETTY IMAGES

Because the Chinese government has revealed little and He is not talking, key questions about his actions are hard to answer. Many of his colleagues and confidants also ignored *Science*'s requests for interviews. But Ryan Ferrell, a public relations specialist He hired, has cataloged five dozen people who were not part of the study but knew or suspected what He was doing before it became public. Ferrell calls it He's circle of trust.

That circle included leading scientists—among them a Nobel laureate—in China and the United States, business executives, an entrepreneur connected to venture capitalists, authors of the NASEM report, a controversial U.S. IVF specialist who discussed opening a gene-editing clinic with He, and at least one Chinese politician. "He had an awful lot of company to be called a ‘rogue,'" says geneticist George Church, a CRISPR pioneer at Harvard University who was not in the circle of trust and is one of the few scientists to defend at least some aspects of He's experiment.

Some people sharply criticized He when he brought them into the circle; others appear to have welcomed his plans or did nothing. Several went out of their way to distance themselves from He after the furor erupted. For example, the two onlookers in that informed consent meeting were Michael Deem, He's Ph.D. adviser at Rice University in Houston, Texas, and Yu Jun, a member of the Chinese Academy of Sciences (CAS) and co-founder of the Beijing Genomics Institute, the famed DNA sequencing company in Shenzhen. Deem remains under investigation by Rice for his role in the experiment and would not speak with *Science*. In a carefully worded statement, Deem's lawyers later said he "did not meet the parents of the reported *CCR5*-edited children, or anyone else whose embryos were edited." But earlier, Deem cooperated with the Associated Press (AP) for its exclusive story revealing the birth of the babies, which reported that Deem was "present in China when potential participants gave their consent and that he ‘absolutely' thinks they were able to understand the risks."

Yu, who works at CAS's Beijing Institute of Genomics, acknowledges attending the informed consent meeting with Deem, but he told *Science* he did not know that He planned to implant gene-edited embryos. "Deem and I were chatting about something else," says Yu, who has sequenced the genomes of humans, rice, silkworms, and date palms. "What was happening in the room was not my business, and that's my personality: If it's not my business, I pay very little attention."

A screen grab from a video shot at the first informed consent meeting for He Jiankui's embryo-editing trial shows Yu Jun (left) and Michael Deem, who were sitting across the table from He and potential participants.

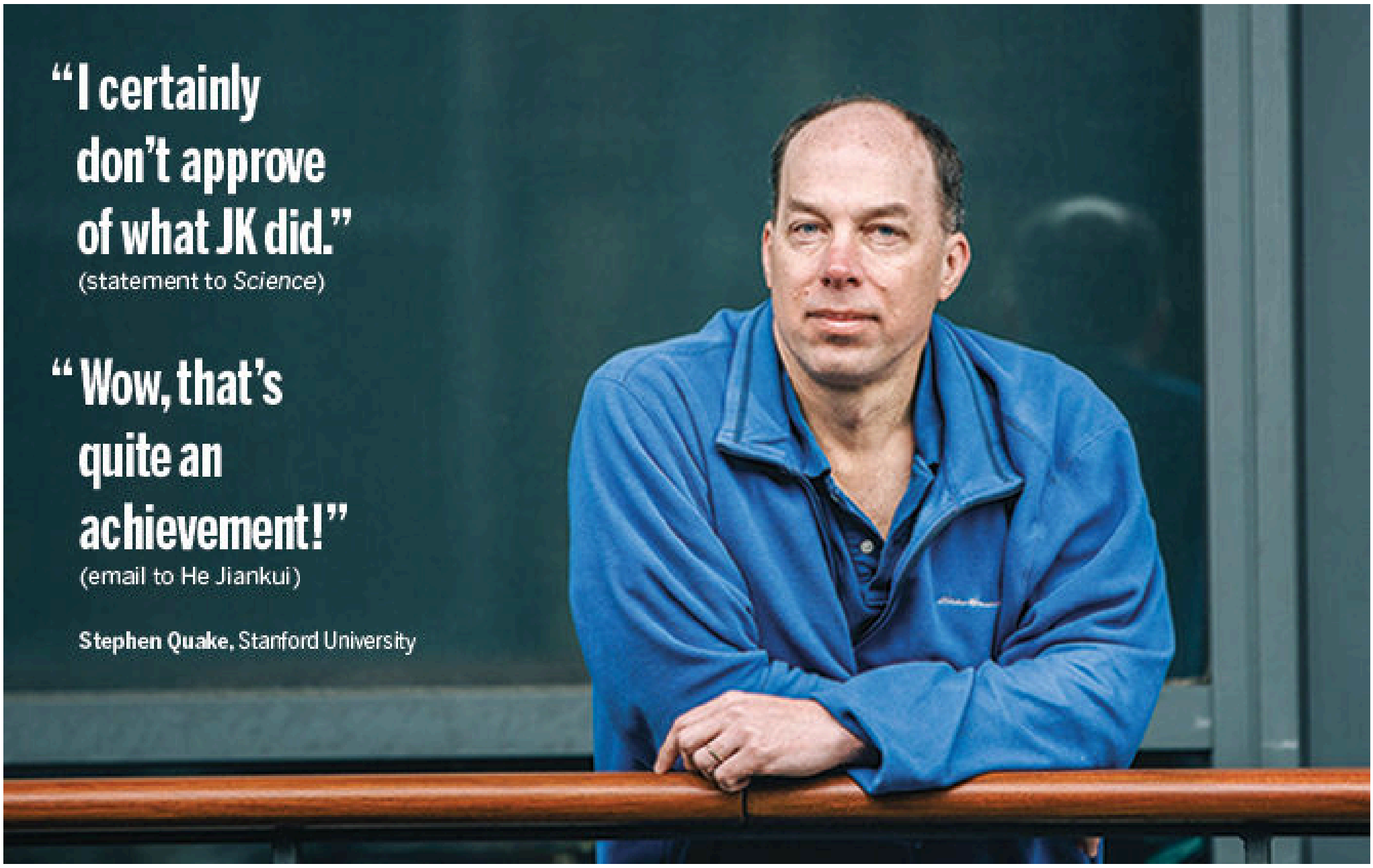
Some people who know He and have spoken to *Science* contend it is time for a more open discussion of how the biophysicist formed his circle of confidants and how the larger circle of trust—the one between the scientific community and the public—broke down. Bioethicist William Hurlbut at Stanford University in Palo Alto, California, who knew He wanted to conduct the embryo-editing experiment and tried to dissuade him, says that He was "thrown under the bus" by many people who once supported him. "Everyone ran for the exits, in both the U.S. and China. I think everybody would do better if they would just openly admit what they knew and what they did, and then collectively say, ‘Well, people weren't clear what to do. We should all admit this is an unfamiliar terrain.'"

A peacock comes home

He, born in 1984, was raised in a village in Xinhua, a county in Hunan province. In high school, according to a Chinese media account, his farming family was so strapped for cash that He repeatedly visited a local bookstore to read a textbook he could not afford to buy. "He still shows a lot of those roots," Ferrell says. "He doesn't that much care for the trappings of wealth. I don't think that was something that motivated him other than that money could be a way to effect change." But by the time He became the subject of international news, his entrepreneurial savvy, honed in the United States, had brought him wealth—and ties to prominent scientists.

Scholarships helped He earn an undergraduate physics degree at the University of Science and Technology of China in Hefei, which explicitly modeled itself after Caltech. In 2007, like many top-notch Chinese graduate students, He went to the United States, joining Deem's lab to work on a Ph.D. in biophysics. They jointly published a mathematical model in 2010 in *Physical Review Letters* that explored evolutionary properties of the CRISPR bacterial immune system, which they wrote "has begun to attract a large amount of attention." Other papers from them analyzed the world trade network, evolution of animal body plans, and influenza genetic sequences.

Deem introduced his student to Stephen Quake, a sequencing guru at Stanford, where He moved for a postdoc in 2011. That same year, He responded to a recruitment ad on the website for SUSTech, a new, well-funded university in the booming city of Shenzhen. The school had a reformer president, chemist Zhu Qingshi—head of He's undergraduate school—who wanted to emphasize high-quality research, creativity, and entrepreneurship. Zhu met with He on a recruiting tour of the United States. In late 2011, on a blog He had started on a website for Chinese scientists, the young investigator announced that he, SUSTech, and Deem would start a joint laboratory to find disease-specific genes that control immune responses.



ANASTASIIA SAPON/*THE NEW YORK TIMES*/REDUX

With funding from Shenzhen's Peacock Plan, one of several Chinese programs to lure researchers back to the country, He set up shop at SUSTech in 2012. "Shenzhen's generosity in encouraging startups, especially venture capitalists—which is comparable to Silicon Valley—is the main reason that attracted me," He later told the *Beijing Review*. "I am not a professor in the traditional sense. I prefer to be a research-type entrepreneur."

After arriving in Shenzhen, with a reported $6 million in startup funding from the Peacock Plan, He launched Direct Genomics, which licensed a technology Quake invented to sequence single molecules of DNA. The firm Quake had co-founded to commercialize it, Helicos Biosciences, went bankrupt that year; now He wanted to resurrect the technology, with a relatively inexpensive machine, as a diagnostic tool for cancer or genetic abnormalities in embryos.

He's new company would lay the foundation for his circle of trust.

Bill Efcavitch, then head of R&D at Helicos, recalls Quake's enthusiasm about He's plans for Direct Genomics. "Bill, this has a chance to bring the phoenix back from the ashes," he remembers Quake saying. Efcavitch, now chief scientific officer of Molecular Assemblies in San Diego, California, immediately liked He, whom he describes as high energy, extremely intelligent, articulate, and funny. "He took the best entrepreneurial spirit that the U.S. has back to China," Efcavitch says. "I didn't think [his company's plan] was a slam dunk, but he had the passion for it."

Efcavitch and Deem joined the scientific advisory board of Direct Genomics, as did Yu, the Chinese genome scientist at the consent meeting. According to a company release, Quake did, too. Quake, who would only give *Science* a statement, denies being on the board.

JK had no intention of blending his interest in doing embryo editing with Direct Genomics, says Ferrell, who was then working for HDMZ, a PR firm the scientist hired in 2015. "He thought it would actually harm the business," says Ferrell, who was not part of the circle of trust himself until after the twin pregnancy had occurred. But He shared his plans with many scientists he met through Direct Genomics.

Efcavitch was an early confidant. Though Efcavitch stresses that he is not a medical doctor, he doubted that any pregnancies would be viable. Efcavitch also thought the experiment could harm He's reputation. "I questioned him, 'Why do you feel you need to take that kind of risk with your personal career?'" Quake similarly knew about the embryo-editing plans early on. In his

statement to *Science*, and in an exclusive interview with *The New York Times* this spring, Quake said he discouraged JK from pursuing the experiment and, when he saw his former postdoc ignoring his advice, cautioned him to make sure he went through the proper ethical and regulatory processes. "I certainly don't approve of what JK did," said Quake, who is co-president of the $600 million Chan Zuckerberg Biohub, in his statement.

One of He Jiankui's colleagues works with a human embryo. MARK SCHIEFELBEIN/AP PHOTO

In April 2018, He told Quake, Efcavitch, and several others linked to Direct Genomics that a woman was pregnant with CRISPR-edited embryos. "I was sort of secretly hoping that the best outcome would be if it failed," Efcavitch says. "I thought the probability of him being demonized was much higher than him being lionized." Quake, *The New York Times* first reported and *Science* confirmed, emailed He, "Wow, that's quite an achievement! Hopefully she will carry to term ..."

Another member of the company's scientific advisory board, Nobel laureate Craig Mello of the University of Massachusetts Medical School in Worcester, reacted differently. Mello told *Science* that he was asked to consult for Direct Genomics by one of its major investors, Li Weibo, a Shenzhen-based philanthropist who funded an institute for rare diseases and endowed a chair at Mello's school. When He emailed news of the pregnancy, Mello replied, "I'm glad for you, but I'd rather not be kept in the loop on this."

Mello, who had co-discovered the gene-silencing process called RNA interference, thought that modifying *CCR5* did not address "a true unmet medical need" and warned He that the experiment was "risking the health of the child you are editing." Mello also noted that it could undermine He's stated goal. "In fact, the treatment itself is feeding fear about HIV and a stigma that is not based on any medical facts," Mello wrote. "I just don't see why you are doing this."

Steve Lombardi, a former CEO of Helicos, reacted far more charitably. Lombardi, who runs a consulting business in Bridgewater, Connecticut, says Quake introduced him to He to help find investors for Direct Genomics. "He's your classic, incredibly bright, naïve entrepreneur—I run into them all the time," Lombardi says. "He had the right instincts for what to do in China and just didn't know how to do it. So I put him in front of as many people as I could." Lombardi says He told him about his embryo-editing ambitions in August 2017, asking whether Lombardi could find investors for a new company that focused on "genetic medical tourism" and was based in China or, because of a potentially friendlier regulatory climate, Thailand. "I kept saying to him, 'You know, you've got to deal with the ethics of this and be really sure that you know what you're doing.'"

Lombardi deemed the CRISPR technology too crude to do embryo editing safely at the time, but he also thought such germline manipulation was inevitable. And he was intrigued by JK's vision of first showing it worked in *CCR5* and then moving on to genes linked to cardiovascular disease and other conditions. Lombardi spoke with several potential investors interested in hearing JK's pitch. "I would have loved to have gone to that half-dozen venture groups and private equity groups," Lombardi says. But in January 2018, He backed out of the meeting Lombardi was setting up. "I just thought, 'OK, it's probably early, and he probably wants to make a splash,'" Lombardi says. "It wasn't the right sort of splash, but he made one."

In April 2018, He asked Ferrell to handle his media full time. Ferrell was a good fit—he had an undergraduate degree in neuroscience, had spent a year in Beijing studying Chinese, and had helped another company using a pre-CRISPR genome editor. Now that a woman in the trial was pregnant, Ferrell says, He's "understanding of the gravity of what he had done increased." Ferrell had misgivings about the experiment, but he quit HDMZ and that August moved to Shenzhen. With the pregnancy already underway, Ferrell reasoned, "It was going to be the biggest science story of that week or longer, no matter what I did."

He Jiankui in July 2017 at Direct Genomics, a company in Shenzhen, China, that he launched after returning to the country 5 years earlier. VCG/GETTY IMAGES

The Doudna connection

On the morning of 26 November 2018 in Shenzhen, the day before the genome-editing summit was set to begin in Hong Kong, "I was in sheer dread, aghast," says Ferrell, who dubbed the date black Monday. The babies had been born the month before, and Ferrell had helped He lay out a plan to go public a month or two after the summit, syncing a published paper with an exclusive given to an AP reporter Ferrell had worked with in the past. But the scheme had unraveled: "This was everything not to plan."

MIT Technology Review had broken a story early that morning China time, saying human embryos were being edited and implanted, after reporter Antonio Regalado discovered descriptions of the project that He had posted online, without Ferrell's knowledge, in an official Chinese clinical trial registry. Now, He gave AP the green light to post a detailed account, which revealed that twin girls—whom He, to protect their identifies, named Lulu and Nana—had been born. Ferrell and He also posted five unfinished YouTube videos explaining and justifying the unprecedented experiment.

"He was fearful that he'd be unable to communicate to the press and the onslaught in a way that would be in any way manageable for him," Ferrell says. One video tried to forestall eugenics accusations, with He rejecting goals such as enhancing intelligence, changing skin color, and increasing sports performance as "not love." Still, the group knew it had lost control of the news.

During the 90-minute drive from Shenzhen to Hong Kong, He desperately sought advice from yet another member of his circle of trust: Jennifer Doudna, the University of California, Berkeley, researcher whose lab in June 2012 had published the first paper showing how CRISPR from bacteria could become a versatile genome editor.

"The nanosecond I landed at the airport, I had just a ton of emails from JK, desperate: I have to talk to you right now, things have really gotten out of control," recalls Doudna, who had played a central, if unwitting, role, in elevating He's status in the CRISPR research community and was a member of the summit's organizing committee. She went to Le Méridien hotel, where He was also staying, and checked in without immediately replying. "He actually had somebody come and pound on my hotel door."

Although Doudna had known He for more than 2 years, she had only learned of his experiment earlier that week when he sent her an email announcing the birth with an attached manuscript. "Honestly, I thought, ‘This is fake, right? This is a joke,'" she recalls. "‘Babies born.' Who puts that in a subject line of an email of that kind of import? It just seemed shocking, in a crazy, almost comedic, way."

Doudna had met He in August 2016 at the annual CRISPR meeting held at Cold Spring Harbor Laboratory (CSHL) in New York. He took a selfie of the two of them and posted it on his blog. In January 2017, He—who that month got a professional boost when he became part of the Chinese government's prestigious Thousand Talents Program—emailed Doudna and asked whether they could get together during his upcoming visit to the San Francisco Bay area. "I am working on the technology to improve the efficacy and safety of genome editing human embryos in China," he wrote.

Coincidentally, Doudna and Hurlbut had organized a small meeting around the time of He's proposed visit to discuss "the ethics and societal aspects of gene editing." Funded by the John Templeton Foundation, the gathering would assemble scientists and clinicians. "We would be delighted to have your participation," she wrote.

Although the participants had agreed not to reveal who attended the meeting, He blogged about it. He wrote that Church was there, as was "our famous stem cell scholar" Pei Duanqing, who directs the Guangzhou Institute of Biomedicine and Health, a CAS branch, and was an author of the NASEM report on human gene editing that would come out a few weeks later. "There were very many thorny questions, triggering heated debates, and the smell of gunpowder was in the air," He wrote of the presentations.

He's own talk, "Safety of Human Gene Embryo Editing," discussed his *CCR5* work in mice, monkeys, and nonviable human embryos discarded from IVF. He stressed that many problems existed with CRISPR, including potentially dangerous DNA cuts inadvertently made at the wrong place in the genome. "Before having resolved these important safety issues, any editing of reproductive cells or creation of gene-edited humans would be extremely irresponsible behavior," he concluded.

ISAAC LAWRENCE/AFP/GETTY IMAGES

To Doudna, JK's talk was something of a shrug, especially because a few groups in China had already reported editing nonviable human embryos—including a team that had published a study that described editing *CCR5* with CRISPR. "I had no inkling that he planned to actually use [CRISPR] clinically," she says. "This was not a person who was publishing much, or even maybe anything in the field, in peer-reviewed journals."

While meeting leaders in the CRISPR world, He was making fast progress on the kind of experiment many of them feared. On 7 March 2017, 5 weeks after the California gathering, He submitted a medical ethics approval application to the Shenzhen HarMoniCare Women and Children's Hospital that outlined the planned *CCR5* edit of human embryos. The babies, it claimed, would be resistant to HIV as well as to smallpox and cholera. (The natural *CCR5* mutation may have been selected for because it helps carriers survive smallpox and plague, some studies suggest—but they don't mention cholera.) "This is going to be a great science and medicine achievement ever since the IVF technology which was awarded the Nobel Prize in 2010, and will also bring hope to numerous genetic disease patients," the application says. Seven people on the ethics committee, chaired by Lin Zhitong—a one-time Direct Genomics director and a HarMoniCare administrator—signed the application, indicating they approved it.

That month, He began to work with Baihualin, a Beijing AIDS advocacy group, to recruit potential participants. Three months later, He met with Hurlbut at Stanford, but did not tell him then that the study was underway. Hurlbut, a physician and neurobiologist who served on the President's Council on Bioethics from 2002 to 2009, saw He as a "nice person" who needed some guidance. "We train these people, and especially foreign postdocs, and yet we don't provide any real context and training on the ethical applications of what they're working on," Hurlbut now says.

Back in Shenzhen, He held the first informed consent meeting 5 days after the Stanford visit.

At CSHL's annual CRISPR meeting the next month, He gave an update on his *CCR5* editing, finishing with a slide of a 1999 *New York Times Magazine* article, "The Biotech Death of Jesse Gelsinger," which recounted a famous gene therapy tragedy. "Gene editing for human germline is going to happen in the near future, so I want to remind everyone we should do this slow and with caution, because a single case of failure may kill the entire field," He said. Again, he did not mention the trial already underway.

On his acknowledgment slide, He listed Hurlbut. "He had, at the point, one deep conversation with me," Hurlbut says. "He perhaps just associated himself with people to gain the credibility that comes with association."

In January 2018, after a first, failed attempt to transfer an edited embryo, He, accompanied by a lab member, returned to Stanford and met with Matthew Porteus, a pediatric oncologist who had attended the Templeton gathering and who helped write the 2017 NASEM genome-editing report. When He told Porteus that an ethics panel had approved his plans for his germline-editing study, Porteus was angry. "I spent the next half-hour, 45 minutes telling them about all the reasons that was wrong, that there was no medical justification; he was not addressing an unmet medical need that, you know, he had not talked about this publicly."

"I left it at that, assuming that I had dissuaded him," Porteus adds. JK sat silently, his face flushed. "[He was] clearly upset, and I guess, as I reflect back, probably somewhat surprised that somebody was giving him such powerful negative feedback."

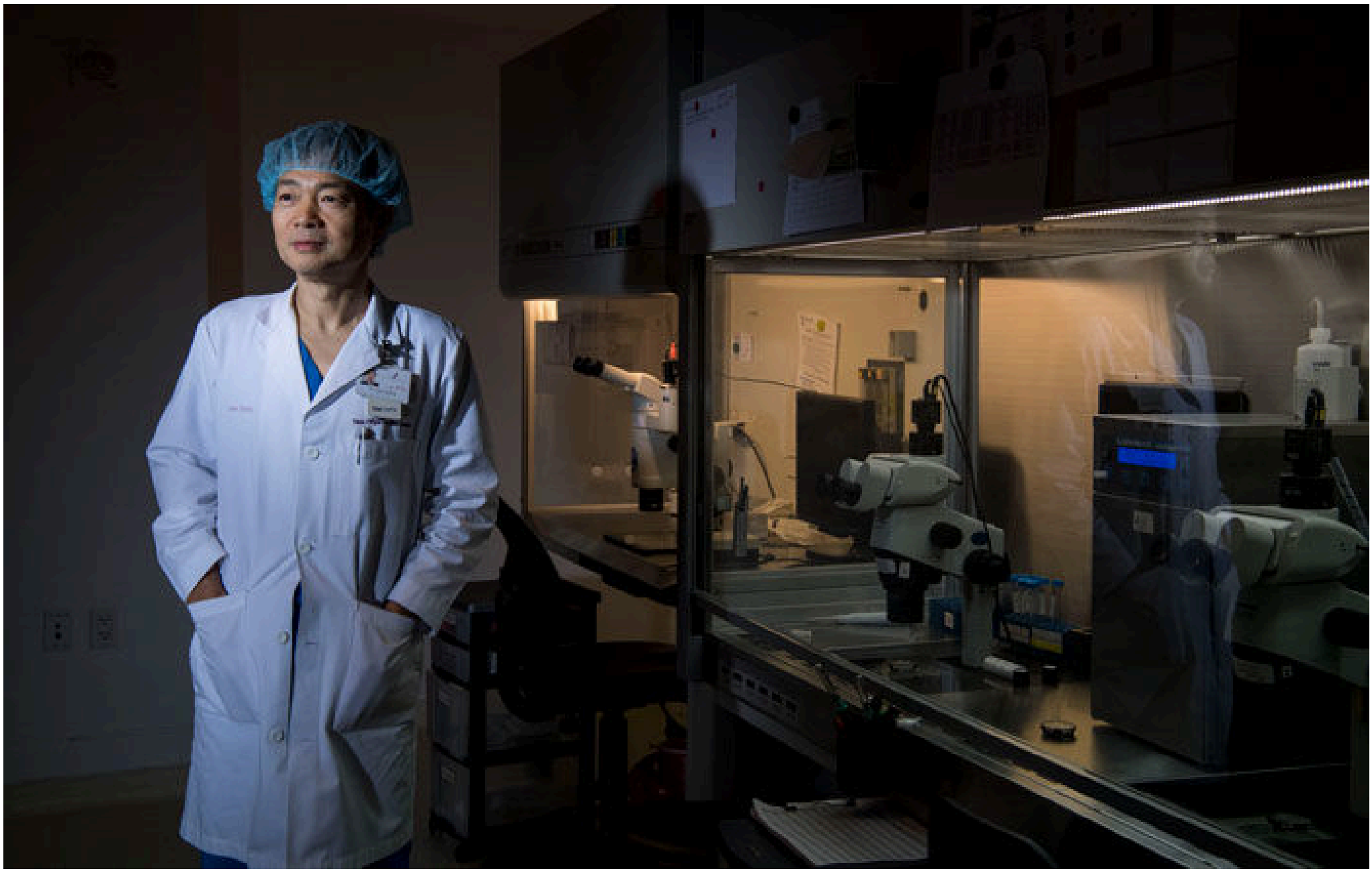
Hurlbut "really admonished him" during the same visit but didn't get angry, despite qualms about destroying embryos in research, let alone germline editing. Hurlbut had developed a strong personal concern for JK. "I felt like I was building a relationship that might preempt foolish behavior," Hurlbut says. "I was intentionally trying to slow him down and influence his thinking."

What neither Hurlbut nor Porteus realized was that He was seeking counsel after the fact. The trial was well underway, and He was thinking about building on his feat.

Mixed wisdom

Last year, a joint venture in New York City between Mixed Wisdom and New Hope Fertility Center launched an ambitious campaign to attract Chinese couples seeking IVF who wanted the latest reproductive technologies. "The future of a nation is in its youth. It starts with genes. To ensure that every baby born in the world is an angel without regret," says an advertising video for the venture, which boasts "one of the busiest infertility centers in the United States."

The center is headed by John Zhang, who earned his medical degree in China and a Ph.D. in reproductive biology at the University of Cambridge in the United Kingdom. Zhang had made international headlines himself in September 2016, when *New Scientist* revealed that he had created the world's first "three-parent baby" by using mitochondrial DNA from a donor egg to revitalize the egg of a woman with infertility and then inseminating the resulting egg. "This technology holds great hope for ladies with advanced maternal age to have their own children with their own eggs," Zhang explains in the center's promotional video, which alternates between Chinese and English. It does not mention that Zhang did the IVF experiment in Mexico because it is not now allowed in the United States.



John Zhang, shown here at his embryology lab at the New Hope Fertility Center in New York City in December 2017, discussed opening a clinic with He Jiankui in China. CAROLYN VAN HOUTEN/*THE WASHINGTON POST*/GETTY IMAGES

Zhang's center also specializes in preimplantation genetic testing, which screens IVF embryos for known DNA mutations linked to disease so that only the healthiest ones can be implanted. "More want to witness our people really use genes to make the country strong," the video concludes. Zhang has argued for pushing the boundaries of reproductive medicine even further. During an interview for a May 2018 article in *The Washington Post*, he reportedly pounded the table in frustration about the controversy surrounding mitochondrial replacement therapy and designer babies in general, noting that people "morally and ethically" accept altering their physical future with plastic surgery. "Zhang argued that gene editing of babies one day will be as safe and common as mobile phones," the article said.

In August 2018, He and Zhang met at Zhang's New York City center to discuss opening a clinic together in China. Zhang, a source says, later met with He in several Chinese cities, including an October trip to Hainan, where they discussed their vision with provincial officials. That tropical island in the south of the country, according the state-run *China Daily*, has plans to become "a world-class medical tourism hub."

When *Science* contacted Zhang, the physician initially said he barely knew He: "I know him just like many people know him, in an academic meeting."

After being presented with details about their interactions, Zhang confirmed that they had a "scientific brainstorm to see what kind of a technique can be developed for the future." He described it as "more of a purely academic discussion" to see how they could "advance the field." Zhang stressed that he had nothing to do with the gene-edited babies and thought CRISPR was still too unreliable for such work. "It's more about how to find a new way to make the procedure more efficient, the techniques to apply in the future clinical use," he said. Zhang conceded that the pair discussed "the potential" of setting up a clinic together. But now, Zhang said, "after all this happened, of course we're not going to have any further collaboration."

The summit

When Doudna finally sat down with He in the hotel lobby on the morning of 26 November, a few hours after the news of the babies broke, the Chinese biologist seemed surprised by the immediate, intense flood of attention and mounting criticism, she recalls. He even asked her whether he should discuss the gene-edited babies in his talk. "It was bizarre," she says. "He seemed so naïve."

For weeks before the news broke, He had wrestled with what to reveal, and when. In a text message obtained by *Science*, He said he had discussed the timing with his wife and with Zhang, a lab member, and "Mayor Xie"—who two He intimates say is Xie Bingwen, a deputy mayor and director of science and technology in Shenzhen's Nanshan district. (Xie did not respond to several emails or calls seeking comment.) "I decided to set the announcement date of birth at around Nov. 20," He wrote. That alarmed Ferrell and the lab, and at their request Quake intervened, as he explained to *The New York Times*, trying to persuade He to wait until he published a paper.

At a dinner with He later on 26 November, Doudna and other summit organizers lobbied for full disclosure. He agreed to describe his work in detail at his talk 2 days later, although he said he had received threatening text messages and had switched hotels for safety. Alta Charo, a bioethicist at the University of Wisconsin Law School in Madison, asked He whether he understood the importance of the principles spelled out in the two main documents that gave germline editing a yellow light of sorts: the 2017 NASEM report and a similar July 2018 report by the Nuffield Council on Bioethics in the United Kingdom.

"I absolutely feel like I complied with all the criteria," He said.

"That kind of rocked me back," Charo says.

The press gaggle at a genome-editing summit made so much noise that the chairperson of He Jiankui's session threatened to cancel it if they didn't quiet down.
NATIONAL ACADEMIES/FLICKR (CC BY-NC-SA 2.0)

By the dinner's end, Doudna says, He seemed upset. If he was reeling from that discussion, it was nothing compared with the public lashing He took before his talk. SUSTech issued a statement saying it was "unaware of the research project and its nature" and a biology department committee concluded He had "seriously violated academic ethics and codes of conduct." HarMoniCare said its hospital "never participated in any clinical operation related to the 'gene-edited' babies incident," the babies weren't born there, no "relevant meeting of the Medical Ethics Committee of the hospital in fact took place," and the signatures on the approval "are suspected to have been forged." More than 100 Chinese scientists signed an online declaration calling the research "crazy."

By the day of He's scheduled talk, the organizers still worried he might not show, and tension filled the University of Hong Kong's Grand Hall in anticipation of his presentation. "I have the right to just cancel the session if there's too much noise and interruption," said session chair Robin Lovell-Badge of the Francis Crick Institute in London at the outset.

When He took the stage, Lovell-Badge had to sharply reprimand the noisy press corps taking pictures. The room calmed down, and He gave a composed, 20-minute talk, followed by a 40-minute question-and-answer period led by Lovell-Badge and Porteus. Scientists in the audience—and more than 1 million people watching a live webcast—strained to analyze He's data slides showing that the gene edits had taken place in one baby and been on target. The other girl, He noted, did not have the edit in both parental genes and thus would not be protected from HIV. His most startling revelation was that another woman was pregnant with a CRISPR-edited baby.

After his talk, He immediately drove back to Shenzhen, and his circle of trust began to disintegrate. He has not spoken publicly since. "I don't think he can recover himself through PR," says Ferrell, who no longer works for He but recently started to do part-time work for He's wife. "He has to do other service to the world."

He Jiankui, flanked by Robin Lovell-Badge (left) and Matthew Porteus, took audience questions at a genome-editing summit in Hong Kong, China. AP PHOTO/KIN CHEUNG

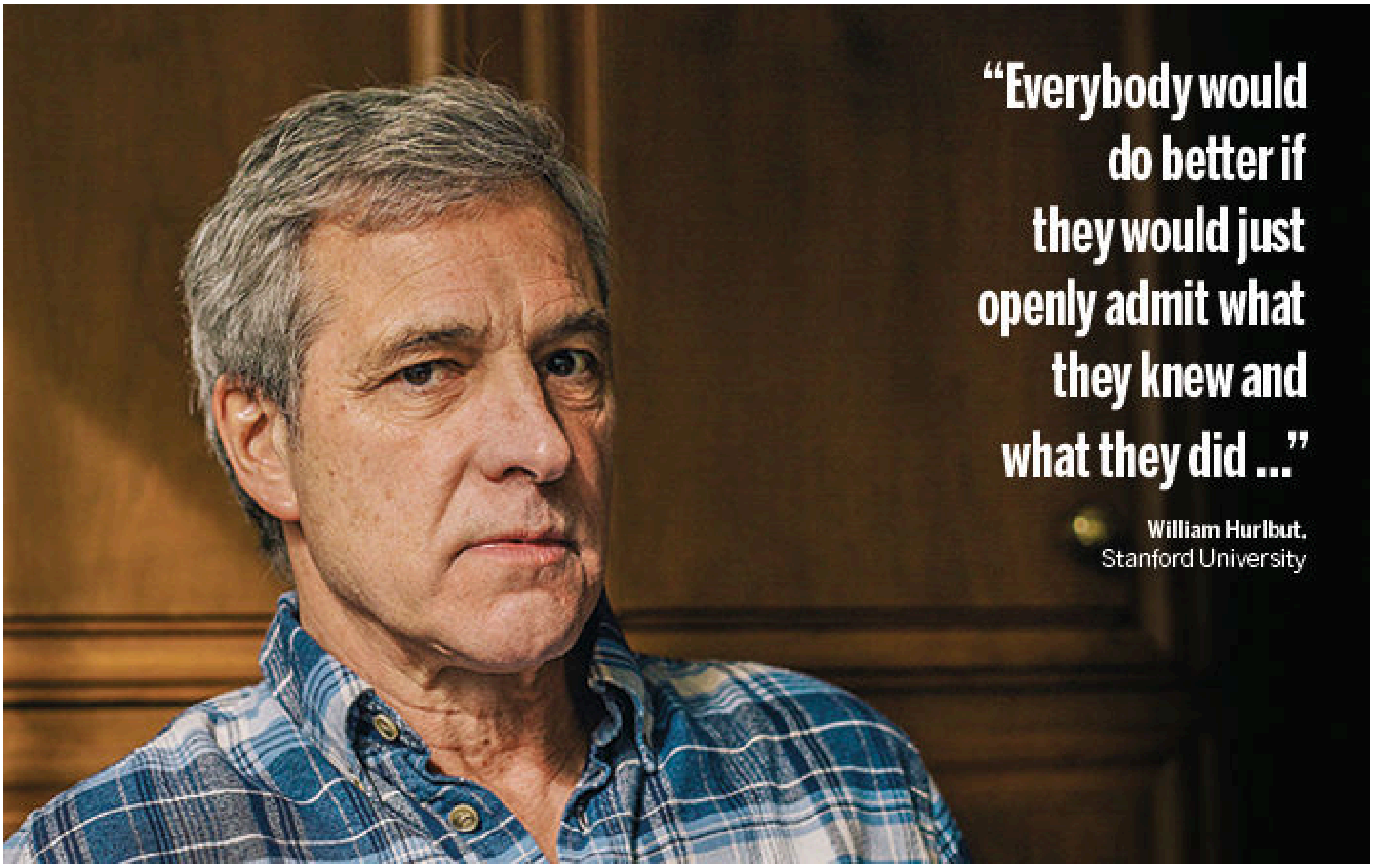
Finger pointing, soul searching

After He's return to Shenzhen, stories suggested he was "missing," under house arrest, or facing the death penalty. In January, He assured Hurlbut in phone conversations that he was in no imminent danger and, although he had guards outside his university apartment, could move freely about campus. An email to Hurlbut also made clear that He had done some soul searching. "I recognize I pushed too quickly into a first-of-kind clinical study without the necessary open dialog with regulators, the scientific community, and the public," He wrote.

The institutional response in China has been feeble. The central government or other bodies have yet to conduct the transparent investigation many people had wanted. SUSTech declared in its one statement on the matter that it wanted "international experts to form an independent committee to investigate this incident, and to release the results to the public." But no such inquiry has occurred.

The only investigation yet made public was one announced by the state-run Xinhua News Agency in Guangdong province, where SUSTech is located. The report provided no documentation and did not name the investigators. "He Jiankui had defied government bans and conducted the research in the pursuit of personal fame and gain," the story states. Punishment would be in accord with unspecified "laws and regulations," and anyone "suspected of committing crimes will be transferred to the public security department."

Among those in He's circle who spoke with *Science*, none reports directly communicating with him after the Xinhua story. No information has been provided about the health of the twins, and the third baby's fate has not been made public.



ANASTASIIA SAPON/*THE NEW YORK TIMES*

Calls for a moratorium on human germline editing have increased, although at the end of the Hong Kong summit, the organizing committee declined in its consensus to call for a ban. China has stiffened its regulations on work with human embryos, and Chinese bioethicists in a *Nature* editorial about the incident urged the country to confront "the eugenic thinking that has persisted among a small proportion of Chinese scholars."

Church, who has many CRISPR collaborations in China, finds it inconceivable that He's work surprised the Chinese government. China has "the best surveillance system in the world," he says. "I conclude that they were totally aware of what he was doing at every step of the way, especially because he wasn't particularly secretive about it."

Some supporters of He in China have abandoned him or gone silent. Zhu, SUSTech's former president who had recruited He but retired in 2014, said initially the work was "very meaningful," according to Sina news, but he later backpedaled, explaining that genetic editing was not his field. *Science* could not reach him.

Pei—the stem cell scientist He met at the Templeton meeting who co-organized the Hong Kong summit and co-wrote the NASEM report—also became entangled in the secret. According to two sources who did not want to be named, He said he informed Pei a few months before the summit about the implanted, edited embryos—which Pei strongly criticized. Pei declined to discuss the matter, saying He's actions were the subject of an ongoing investigation.

SUSTech's current president, who succeeded Zhu, has tried to shift blame in letters to Stanford, saying Quake "provided instructions to the preparation and implementation of the experiment, the publication of papers, the promotion and news release, and the strategies to react after the news release." After conducting "a review" of He's interactions with Quake, Hurlbut, and Porteus, Stanford exonerated all three researchers—without releasing any details of its probe.

Benjamin Hurlbut, William's son and a historian of biomedicine at Arizona State University in Tempe, says leaders in the scientific community should take a hard look at their actions, too. He thinks the 2017 NASEM report helped give rise to He by following a well-established approach to guiding science: appointing an elite group to decide how scientists should be regulated. Benjamin Hurlbut, whose book *Experiments in Democracy* explores the governance of embryo research and bioethics,

questions why small, scientist-led groups—à la the totemic Asilomar conference held in 1975 to discuss the future of recombinant DNA research—are seen as the best way to shape thinking about new technologies. Hurlbut has called for a "global observatory for gene editing" to convene meetings with diverse perspectives.

The prevailing notion that the scientific community simply "failed to see the rogue among the responsible," Hurlbut says, is a convenient narrative for those scientific leaders and inhibits their ability to learn from such failures. "It puts them on the right side of history," he says. They failed to paint a bright enough red line, Hurlbut contends. "They are not on the right side of history because they contributed to this."

Several authors of the NASEM report reject that charge, stressing that He ignored their strict criteria and that they had called for government regulation of human germline editing in addition to self-regulation by the scientific community. "It is straining the facts to assert that anything in the report gave any license to JK," says one author, Richard Hynes of the Massachusetts Institute of Technology in Cambridge. "So, did self-regulation work? Not in this case. That does not mean that self-regulation is inappropriate or unnecessary. In fact, it should be an important adjunct to governmental regulation."

"Can one truly believe that He Jiankui would have decided not to proceed, simply because our committee had advocated a moratorium or other kind of prohibition?" Charo asks. Two international committees, including one under the World Health Organization, have formed to debate creating a sturdier framework to govern and regulate germline editing.

Doudna suggests that germline editing became inevitable in 2015, before He even began his experiments. "That decision was made when the first person started working with human embryos using CRISPR," Doudna says. "To me, the question now is, 'How will it happen? How do we control, or can we even control it in any reasonable sense of that word? And what will the impact be on society going forward?'"

***Clarification, 1 August, 1:05 p.m.:** This story has been updated to note Ryan Ferrell's current part-time employment with He Jiankui's wife.*

***Correction, 2 August, 11:15 a.m.:** This story has been updated to reflect that Yu Jun co-founded the Beijing Genomics Institute, not BGI Genomics.*

RELATED STORY

The long shadow of a CRISPR scandal

BY JON COHEN

handbag

As He Jiankui strode to the podium at last year's summit on human genome editing in Hong Kong, China, more than 1 million people watched online. ANTHONY WALLACE/AFP/GETTY IMAGES

In the months after He Jiankui's widely condemned embryo editing went public, Chinese researchers using the genome editor CRISPR reeled with embarrassment, outrage, and fear of unwarranted scrutiny and criticism of their own work. Some see He as a nobody in the country's CRISPR community who sullied their whole field. "Many Chinese scientists got angry about [He]," says Deng Hongkui, a stem cell researcher at Peking University in Beijing. "Many of us got training in Western countries, and we know the international standards."

An acute concern is that the scandal will besmirch China's many studies of CRISPR medical treatments that involve altering the DNA of adult somatic cells, such as skin, neurons, and muscles. Edits to those cells, unlike the germline cells in He's embryo experiments, are not passed to future generations. Many people, Deng says, don't understand the difference between germline and somatic cell work. "They just hear 'genome editing,'" Deng says. He used CRISPR to make blood-forming stem cells resistant to HIV, and then gave them to an HIV-infected patient who also had leukemia and needed the transplant to battle that disease.

Wang Haoyi, a developmental biologist at the Institute of Zoology in Beijing, a branch of the Chinese Academy of Sciences (CAS), resents the image He reinforced of Chinese research as the Wild West. "All the news is that Chinese scientists did this, not that it's a single person—it's like they're crazy people," says Wang, who did pioneering CRISPR mouse studies in a U.S. lab.

Yang Hui, who collaborated with Wang on those mouse studies and now is at CAS's Institute of Neuroscience in Shanghai, says the day after the news broke about the gene-edited babies, he took the unusual step of publicly posting a paper under review at *Science*. It showed that a CRISPR variant called a base editor had substantial, unexpected off-target effects. "We want foreign people to know Chinese scientists also do some good stuff," Yang says. "We want to make safer gene-editing tools for patients."

The He "incident" has forced Chinese researchers to think twice about future projects. "People aren't daring to do some CRISPR stuff because they don't want to be criticized," says Chen Jia, who studies DNA repair at Shanghai Tech University. Chen, who contributed a CRISPR tool used in a study with human embryos never intended for implantation—one of nine such published studies done in China to date—says he would be hesitant to join a similar project today. Even though Chen only provided the gene-editing tool, the He fallout "changes things."

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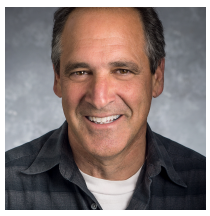
ABOUT THE AUTHOR



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